

A Smart Computing Framework for AI-Driven Urban Drainage and Waterlogging Prediction

Sonam Singh¹, Tushar Chavhan², Sakshi Ingale³, Vrushali Dhage⁴, Snehal Godage⁵, Shubhangi Bhawari⁶

MIT Academy of Engineering, Alandi, Pune, Maharashtra, India¹

Dr. D. Y. Patil Institute of Technology, Pimpri, Pune, Maharashtra, India^{2,3,4,5,6}

¹sonamchauhan346@gmail.com, ²chavhantushar223@gmail.com, ³sakshiingale2212@gmail.com

⁴vrushalidhage19@gmail.com, ⁵sgodage46@gmail.com, ⁶shubhangi.bhawari@dypvp.edu.in

Abstract—There is a continual increase of challenges faced by cities that have grown rapidly as a result of increased rainfall variability, unplanned urbanisation and aging drainage infrastructure. Increasing urban development has created challenges that have made urban flooding an ongoing issue and therefore traditional methods for monitoring drainage and predicting flooding have relied on isolated data sources or intensive computer simulations. To address these issues, this paper presents a framework using Smart technologies that combines Artificial Intelligence (AI) for monitoring urban drainage and Waterlogging Risk Management with IoT for water level sensors, GIS for mapping drainage, and real-time weather data in an integrated operation. The framework highlights a modular edge-cloud integration of multiple sources of data into a decision making tools to provide situational awareness for proactive management of drainage systems. In place of a traditional focus on benchmark testing algorithms, this paper is an example of a systems design approach to urban drainage by presenting, Architectural choices, Data Workflows, and Operational Considerations for Smart City deployment. The framework can help build a foundation for future experimental validation, continuous improvement and deployment of intelligent and climate resilient urban drainage systems to cities.

Index Terms—Urban drainage systems, waterlogging risk management, Internet of Things (IoT), Geographic Information Systems (GIS), smart cities, artificial intelligence (AI), multi-source data fusion, edge-cloud computing, decision support systems.

I. INTRODUCTION

Due in large part to shifting rainfall patterns, fast urbanization, and antiquated drainage systems, urban waterlogging has become an increasingly widespread and enduring problem in cities all over the world [1]. With a large percentage of paved and impermeable surfaces, modern cities are particularly vulnerable; even light rainfall can now result in flooded streets, traffic jams, financial losses, and health hazards for the general public [2]. As climate change brings more heavy and unpredictable rainfall, events that were once thought to be extreme are happening more often [1]. Flooding is exacerbated in many urban areas through overburdened drainage systems, dwindling natural water bodies, and improper waste management.

Despite increasing use of artificial intelligence to forecast flooding, most of the existing systems are inefficient and ineffective due to their heavy reliance on either simulations or a single data source [3]. Conventional water management methods, despite their precision, even though hydrodynamic models are frequently necessitating substantial resources, sluggish, and

reliant on data, therefore unsuitable for quick decision-making in busy metropolitan regions [4]. In the same way, AI models that depend exclusively on Meteorological data and flood records frequently have information deficiencies, a deficiency in spatial information, and an incapacity to adapt to novel or severe climatic conditions [5]. The failure of multiple systems to effectively represent real-world conditions results in the inability to consider the complexity of how water moves through drainage systems or integrate real time information from IoT sensors [6].

The limitations of these systems result in reduced effectiveness in managing flooding and can reduce the overall long-term sustainability of urban areas [7]. Low-income communities with high levels of vulnerability are often disproportionately affected by waterlogged homes, contaminated drinking water and increased rates of disease due to flooding [1]. Employing a multi-faceted approach will allow planners and emergency responders to create urban environments that are adaptable, sustainable, and resilient through providing them with a real-time water level monitoring IoT System and GIS drainage mapping along with real time weather data to Increase the accuracy of Forecasting and improve their decisions in regards to those environments [8] [9].

The key contributions of this paper are as follows:

- A service-oriented AI-based infrastructure supports urban flooding risk evaluation and management by combining IoT-sensors and GIS modeling to provide an instantaneous simulation of hydrology, combined real-time weather metrics.
- A system architecture built on a system architecture between remote servers and local machine in order to provide coordinated data between several different locations, scaling up as needed and providing rapid response and management tools for urban drainage operations.
- A qualitative design-level evaluation detailing how the combination of the above will address the short-comings of the current methods of operating and monitoring drainage systems.

II. LITERATURE REVIEW

A great deal has been achieved in predicting and detecting urban flooding and obstruction in urban drainage systems due to recent improvements in Artificial Intelligence (AI) and Machine Learning (ML) technology [4]. Predictive models such as Random Forests, Long Short-Term Memory (LSTM) Networks, and Ensemble Techniques are currently capable of generating fast and accurate predictions regarding water levels and the likelihood of a flood [7]. Compared with traditional hydraulic simulation-based approaches that often require large amounts of processing time and fixed data sets, these modern AI-based predictive models can quickly adapt to the ever-changing urban environment and its complexities [8]. Additionally, the combination of AI and physical simulation enables near-real-time forecasting of urban floods; allowing for proactive planning of maintenance activities and emergency responses [2]. However, while modern predictive models have many advantages, their predictive accuracy is highly dependent upon the quality, variability, and consistency of the input data [3].

The Internet of Things (IoT) has also created new possibilities for the continuous monitoring and management of Urban Drainage Systems, enabling city managers and engineers with real-time data to make informed decisions [6]. IoT devices that are networked together can monitor water levels, flow rates, debris accumulation, and automatically control pumps and valves to prevent an overflow in case of a storm event. Using Satellites, the City will have the ability to predict heavy rainfall events with unprecedented detail and lead times, and anticipate potential blockages, thus providing early flood warnings when combined with AI Analytics [5]. Applications in the real world have demonstrated that this combination can increase the cost-effectiveness and proactiveness of drainage management [9]. However, the majority of these implementations are still small-scale, with problems like erratic connectivity and challenges when trying to scale to larger, older, and more varied city infrastructures.

In order to comprehend how water flows through cities, Geographic Information Systems (GIS) are also essential [1]. GIS tools are useful for mapping drainage networks, identifying areas that are vulnerable to flooding, and analyzing the interactions between infrastructure, terrain, and land use [10]. GIS enhances spatial accuracy and assists in identifying potential waterlogging locations and times when combined with AI [11]. However, it is still rare that the full integration of GIS with AI analytics and real-time IoT data, which narrows down the possibility of truly comprehensive flood forecasting systems [8].

A multidisciplinary research approach that integrated different data types was used, such as meteorological data, hydrological patterns, spatial layouts, and real-time sensor readings—into unified systems in order to get around these constraints [7]. This model leverages multiple sources of data, compared to models that use one data source. The fusion of source data enables the making of more accurate, real-

time, and context-aware predictions [4]. However, there are still issues in integrating these different data streams in a smooth manner, ensuring the quality of the data, and maintaining interoperability between systems operated by different organizations or built with various technologies.

Despite advancements, a large number of AI-based drainage solutions still depend on discrete datasets, which restrain their ability to adapt and survive in operational situations [12]. There are very few frameworks that successfully combine IoT sensor feedback, GIS-based drainage mapping, and meteorological data onto one single, coherent platform [13]. Scaling and integration are difficult since there are fragmented data infrastructures and a lack of standardized protocols to exchange data [3]. Furthermore, long-term resilience and sustainability aspects such as infrastructure decay, the impacts of climate change, and the need for equitable access to flood protection—must be frequently reconsidered [7]. As cities continue to grow and with ongoing technological development and innovation, the challenges associated with the increasing infrastructural demands of urban drainage will only continue to increase. As urban drainage systems are developed in order to meet these demands; they will also have to be designed in such a way that they will be both agile and smart. In this regard, the literature has illustrated the potential for the use of Artificial Intelligence (AI), Internet of Things (IoT) based sensing and Geographic Information System (GIS) tools to achieve monitoring and assessment of flood risk for urban drainage systems; however; the vast majority of existing research has been limited to isolation of various aspects of urban drainage systems such as data based predictive modelling, sensor based monitoring, and spatial analysis; often treated as individual stand-alone solutions, creating additional barriers to effective inter-operability between system components, real-time data fusion, scalable deployment, and decision support for operational use of data and models therefore these approaches to urban drainage enable conditions to remain limited. In addition, the majority of previously published methods focus on the degree of accuracy for algorithms while neglecting the potential for practical implementation of the methods in urban infrastructure, and lack of adaptive system design abilities. In this paper, a modular and unified framework which integrates sensing, spatial analysis, and intelligent decision support using a coherent edge-cloud architecture is proposed to fill the gaps identified in existing studies for smart city drainage applications.

III. PROPOSED METHODOLOGY

1. Source of Input Data

The framework outlined will make use of multiple heterogeneous data sources to build a complete picture of how urban drainage systems function. Specifically, this framework will incorporate the following types of data to assist with creating an accurate urban drainage system model:

- **Real Time Climate Data:** Real-time climate data is collected from meteorological networks and forecast services, which will provide real-time information about rainfall characteristics [1] to assist with assessing flood risk and performing situational analysis as part of the framework. [5].
- **GIS Network Maps :** High-resolution GIS maps will be developed that depict the entire urban drainage network and its component parts (e.g., pipes, catch basins, manholes, etc.) in relation to each other and with other infrastructure (e.g. roads, terrain) [4].
- **Water Level Sensors:** Constant monitoring of water levels, flow rates, and significant fluctuations will occur through the use of IoT-enabled smart sensors that support staff in identifying and responding to issues related to drainage channel blockages and/or wet soils due to changes in the environment. [13].

2. Architecture of AI Models

Temporal rainfall patterns and Spatial Features of Urban Geography: Users Will Understand How Urban Drainage Systems Function Using AI Analytics Set on Temporal Patterns of Rainfall Analytics and Its Relationship to the Urban Geography.

- **Temporal Analysis:** Users of the technology will utilize the tools provided to consider both temporally as well as spatially how rain fall and water levels change through various hydraulic systems in any area and to relate this to urban geography (i.e., how/where/when/etc). [13].
- **Spatial Analysis:** Users of the technology will be able to access GIS-based derived Spatial Analysis to provide them with GIS-based Derived spatial information of how the geographic features of an urbanized environment impact hydraulic systems, thus providing further understanding of the location, movement, and delay hazards for each individual hydraulic system. [11].
- **Module Integration:** The temporal and spatial aspects of this module are integrated to provide the framework with 4D capabilities for assessing the risk of flooding based on the spatial distribution of rainfall and the drainage infrastructure in various areas of the city [10]. [8].

3. Data Fusion Strategy

The framework integrates data from multiple heterogeneous sources, including sensor measurements, historical rainfall information, and drainage maps, to support a unified data representation for flood-risk assessment within urban drainage systems.

- **Feature-Level Fusion:** The use of feature-level data fusion provides a means to combine, integrate or align multiple diverse data types so that the information can be analyzed within the context of multiple factors (e.g., Environment - ie; climate, hydrology, [6].
- **Hybrid Ensemble Learning:** The Hybrid Ensemble Learning (HEL) architecture utilizes a Conceptual Integrated Framework which combines data-driven AI centric components of Model Output with Hydrologic Models to

create a more Robust means of assessing flood risk by resulting in a higher correlation between Hydrologic Assessments and Drainage System behaviour for a particular time frame or location. [12].

- **Dynamic Model Updates:** The ability to adaptively update Dynamic Models is one of the features of the HEL Framework that allows for the iterative process of updating models based on changes in the way urban infrastructure functions and is inhabited (i.e.; New Developments). [9].

4. Forecasting and Decision-Making Process

The proposed framework operates as a continuous, real-time decision-support pipeline designed to assist urban drainage monitoring and management:

- **Data Acquisition:** Drainage Maps Based on GIS, Data Collected from IoT Sensors, and Real-Time Weather Updates are Collected, Linked Together, and Contained Within a Permanent Repository to Enable the Analysis of the Overall System, and Support for Situational Awareness [13].
- **Preprocessing:** Validation, Cleaning, and Spatially Aligning of the Incoming Data for Consistency and Reliability will be Conducted Using Consistent Reference Framework. The Process will Occur Prior to the Use of the Data in the Analytical Components of the Framework [3].
- **Feature Engineering:** Relevant environmental and infrastructural attributes—such as precipitation levels, historical water observations, elevation, and proximity to drainage infrastructure—are derived to support contextual risk assessment and analytical interpretation within the system [1].
- **Model Inference:** The Artificial Intelligence Analytical Component Will Evaluate the Integrated and Spatially Defined Data Sets to Allow for It to Determine Water Logging Risk within Various Drainage Zones and Urban Regions [1].
- **Decision Support:** The Framework Will Use Visual Dashboards, Alert Symbols, and Contextual Information to Assist Emergency Responders and City Planners With Informed Decisions. Additionally, The Framework Will Provide an Architecture to Support Integration With Automation Drainage Control Where Appropriate [1].
- **Continuous Learning:** The Framework is Set Up to Allow for the Continuous Update of Analytical Components Based on the Latest Available Data and Continue to Adapt to the Changing Environmental Conditions and Infrastructure [6].

Overall, the proposed framework provides a structured, data-driven approach for supporting resilient urban drainage management by integrating sensing, spatial analysis, and AI-based decision support within a unified system design.

5. Block Diagram

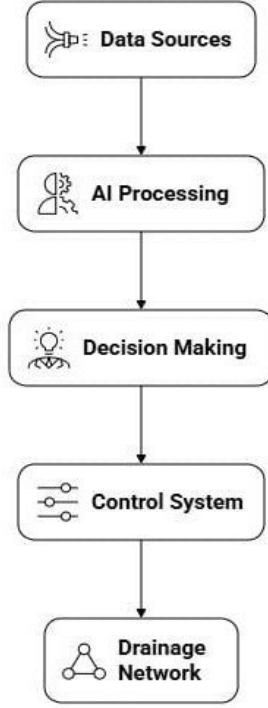


Figure 5.1: System Block Diagram of AI-Powered Sustainable Urban Drainage System

IV. SYSTEM ARCHITECTURE

The proposed AI-based sustainable urban drainage framework adopts a modular, multi-layered architecture that integrates real-time monitoring, analytical processing, and automated control components to support urban waterlogging management [9]. Each architectural layer is designed to perform specific functions while maintaining coordinated interaction with other layers to enable coherent and data-driven system operation.

1. Layer of Data Acquisition

The framework that collects all data from the environment and/or infrastructure is known as the Data Acquisition layer.

- **Meteorological Data Sources:** Data from both meteorological forecasting and meteorological stations (on-going) about how often it rains and its amount and where it rains, as an example (how frequently does rainfall occur; the average length of time it stays; where does rain fall; etc.). [5].
- **IoT Sensor Network:** The use of an IoT Sensor network for real-time monitoring of the flow of water and its level(s) along the drainage network and catch basins, using ultrasonic water level sensors, flowmeters and water quality sensors. [13].
- **GIS Drainage Maps:** High-resolution GIS maps provide accurate and reliable geographical data about the drainage

network, elevation, and land use of urban water management (GIS Maps). [4].

2. Edge Computing Layer

The edge computing layer is designed to support localized data processing and preliminary analysis close to the data sources within the urban drainage infrastructure.

- **Edge Gateways and Microcontrollers:** The edge computing layer leverages edge computing gateways and the microcontrollers like Raspberry Pi and ESP32 to conduct the first layer of localised data processing at the source of the data. This processing includes filtering, normalising and performing basic feature extraction, all of which will save, on the amount of data that must be sent to higher level systems. [3].
- **Local AI Inference:** Localized AI systems based on lightweight, resource-efficient neural network models, have been developed to process data collected by sensors and thus augmented to identify anomalies and patterns in real time using information processed locally. This localized processing of sensor data is intended to improve response time and provide higher-level decision support. [9].

3. Fusion Layer and Centralized Analytics

This layer is designed to integrate heterogeneous data streams and support centralized analytical processing for informed urban drainage management.

- **Cloud Platform:** Centralized cloud infrastructure is used to aggregate meteorological information, preprocessed edge data, and GIS-based spatial data, providing a scalable environment for advanced analytical and simulation-based processing [3].
- **Data Fusion:** The data fusion module integrates historical sensor observations with spatial and meteorological context to support comprehensive assessment of waterlogging risk and drainage behavior across urban regions [7].
- **Decision Support:** The decision-support component provides visualizations, status indicators, and contextual information to assist city officials in informed planning and maintenance-related decision-making. These outputs are intended to support proactive management of urban drainage infrastructure [2].

4. Layer of Control and Actuation

The drainage related Layer of the Urban Infrastructure Response Layer is designed to respond in real-time to System Information when providing control actions and Operational Awareness of Urban Infrastructure in relation to Drainage Systems.

- **Smart Pumps and Valves:** the Smart Pumps and Valves concepts provide a basis for the automation of the operation of Pumps and Valves in response to System Conditions and are aimed at regulating Flow Condition in response to the change of Environmental Scenarios. [6].
- **Dashboard and user interface:** there are Map Oriented User Interfaces built to assist Operators of the System

with situational awareness, providing timely information for informed decisions based upon Water Level Conditions, Areas Sensitive to Drainage, and Indicators of System Conditions. [13].

5. Layer of Feedback and Adaptation

This layer is designed to support long-term system adaptability, scalability, and operational robustness.

- **System Adaptation:** The model provides an opportunity to update the analytical components of the system on an iterative basis by using feedback from observations made of the system as well as additional data streams as they become available. The intent of this iterative update process is to provide the flexibility and adaptability needed to accommodate changes that occur over time to the environment and infrastructure within the area of operation. (Hansen, 2009). [9].
- **Scalability and Resilience:** The architectural structure of the model includes the use of both distributed computing principles as well as redundant methods used for the communication systems (e.g., a LoRaWAN mesh network) to provide a scalable deployment and resilient operations across large municipalities. (Hansen, 2009). [3].

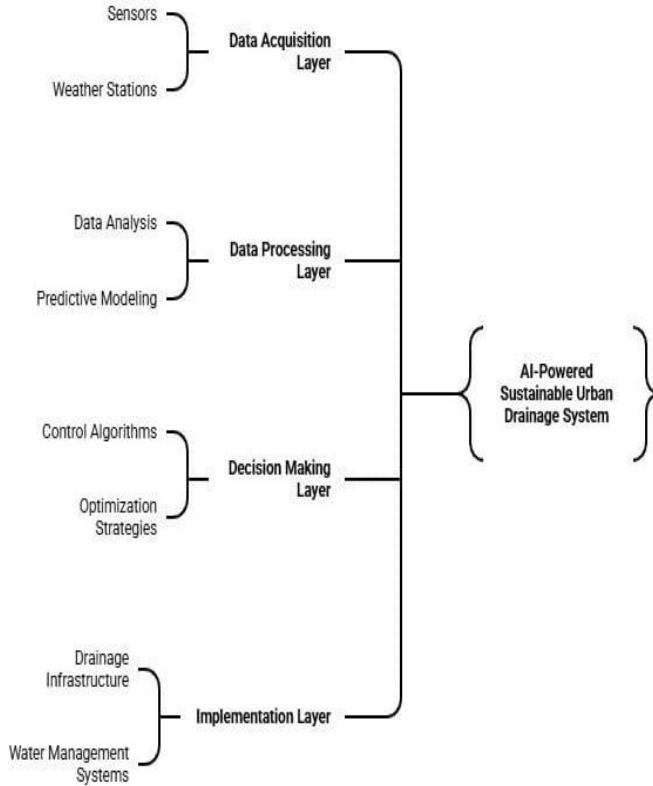


Figure 6.1: Conceptual system architecture illustrating the data flow and functional components of the proposed AI-enabled urban drainage framework

V. QUALITATIVE EVALUATION AND DESIGN DISCUSSION

The proposed multi-source AI-driven urban drainage framework is evaluated qualitatively through a design-oriented analysis, focusing on system architecture, data integration strategy, and operational feasibility rather than quantitative benchmarking. This evaluation approach is appropriate for early-stage system frameworks where the primary contribution lies in architectural integration and workflow design rather than in algorithmic optimization or dataset-specific performance.

The multi-layered edge cloud architecture is a key strength of the proposed framework and provides timely monitoring and decision-making support for urban drainage environments with increasing complexity. By Shifting data preprocessing and preliminary inference to edge devices, the framework aims to reduce the need for centralized processing, enabling local systems to respond more quickly to rapid changes in water level. This approach addresses the challenges of latency and scaling that are present in traditional, central SCADA-based monitoring systems for drainage systems. The proposed framework supports a more complete Catalog of the drainage system due to the inclusion of several different types of data sources including IoT-based water level sensors, GIS-derived spatial drainage information, and real-time meteorological data. While isolated or point solutions monitor and use only one type of data, the framework emphasises the benefit of Data Fusion at the feature level to enable more effective and improved risk assessment and situational awareness. This integrated view of the drainage system is critical in urban environments due to the influence of both the fixed physical infrastructure of the drainage system and the dynamic nature of the weather. From a design standpoint, the framework supports adaptive and extensible operation. The modularity of the components, e.g., sensing infrastructure, analytical models, decision-support tools, allows each of these components to be updated and replaced as well as expanded without a substantial modification of the existing architecture.

The decision support layer adds to the practical usefulness of the framework. The framework is intended to provide city officials and emergency responders with a decision support tool. The framework takes analytical outputs and turns them into visual maps and alert notifications and actionable insights. It is designed to help manage drainage proactively and reduce the risk of flooding. Although the framework conceptually includes automated actuation devices, the deployment and evaluation of the devices will be done after the framework has been proven reliable and safe to use.

While there are many positive features associated with the proposed framework, the authors acknowledge that the framework has a number of limitations. The entire concept behind the framework relies entirely upon three primary elements, all of which need to be present in order for the framework to be successful: data availability, sensor reliability, and reliable communication, especially in urban settings where infrastruc-

ture and population density are high. A final limitation of the framework is that while the framework illustrates how artificial intelligence-based predictive elements can be integrated, the framework does not contain detailed experimental validation, performance benchmarking, or a comparison of the framework's performance against baseline models. Overall, this qualitative evaluation shows that the proposed framework provides a well-defined and scalable system to support the monitoring of urban drainage through the application of artificial intelligence. Additionally, because the proposed framework eliminates the fragmentation associated with sensing, spatial analysis, and decision support, the framework can be used to provide the structure needed for the continued development, experimental validation, and performance-based optimization of AI-enabled urban drainage monitoring.

VI. CONCLUSION

This study has developed a framework for Smart Urban Drainage Monitoring Management and Waterlogging Risks from Multiple Sources using Smart Computing. The Smart Urban Drainage Monitoring Management and Waterlogging Risks using Multi Source Smart Computing framework unites the Internet of Things (IoT) based water-level sensing, Geographic Information Systems (GIS) based spatial information of drainage, and a real time meteorological (weather) monitoring devices within an integrated edge-cloud architecture to mitigate the negative impact of currently fragmented and centrally controlled monitoring systems. The proposed solution was developed to provide support for proactive decision-making for urban drainage monitoring within complex urban environments while being modular, scalable and context aware.

Instead of simply providing an evaluation of the performance of the algorithms (formulaic benchmarking) the work provides a view from the systems level and presents the physical architecture, data flows and considerations around operational deployment necessary to deploy intelligent urban drainage systems. The qualitative analysis presented demonstrates the ability of Multi-Source Data Fusion and Distributed Processing to support improved situational awareness and adaptive Urban Drainage Management for smart cities.

Although the present study does not provide quantitative performance analysis or experimental validation of the proposed framework, it does provide a framework for forthcoming research, the deployment of urban drainage systems in the field, and empirical evaluation. As flooding pressures from climate change variability and urban stresses increase urban flooding risks will require integrated adaptive systems of Urban Water Management to operate effectively, so integrated and adaptive systems will allow for the continued evolution of Urban Water Management.

VII. FUTURE WORK

The future scope of the framework consists of potential improvements, which are going to be the impact on the per-

formance, the scalability, and the reliability of the framework. Remote sensing merged with rainfall as monitored by satellites. On the other hand, federated learning would permit the joint training of models in different cities without breaching their data privacy, whereas nowcasting could foresee the conditions of a larger area and give longer predictions. The powerful protocols like LoRaWAN mesh would be the great choice to allow the reliable data transmission and to strengthen the communication networks particularly during the emergency situations. Digital twins, or the virtual representations of the drainage systems, would be the great option to allow the simulations of different scenarios and the optimization of the systems before the actual implementation in the world. Moreover, the extension of the framework's cloud-edge hybrid architecture and modularity could not only be scalable to heterogeneous infrastructures and megacities, but also, the use of explainable AI techniques to enhance model interpretability would support regulatory acceptance and win stakeholder trust.

REFERENCES

- [1] D. Misra, G. Das, and D. Das, "Artificial intelligent based smart drainage system," 2022, preprint available at Research Square. [Online]. Available: <https://doi.org/10.21203/rs.3.rs-1338435/v1>
- [2] R. Dabas, T. Imam, F. Safwat, S. Rizwan, and K. Alam, "Smart drainage system for urban flood prevention."
- [3] M. M. Saddiqi, W. Zhao, S. Cotterill, and R. K. Dereli, "Smart management of combined sewer overflows: From an ancient technology to artificial intelligence," *Wiley Interdisciplinary Reviews: Water*, vol. 10, no. 3, p. e1635, 2023. [Online]. Available: <https://doi.org/10.1002/wat2.1635>
- [4] S. Boughandjioua, F. Laouacheria, and N. Azizi, "Integrating artificial intelligence for urban drainage systems aided decision: State of the art," *International Journal of Artificial Intelligence: Theory and Applications*, vol. 2, no. 2, pp. 1–9, 2023. [Online]. Available: <https://int-j-aita.univ-annaba.dz/wp-content/themes/cbusiness-investment/view/vol2iss2/vol2iss2-1.pdf>
- [5] D. P. Selvam, G. G. Reddy, and N. M. Kumar, "Designing a smart and safe drainage system using artificial intelligence," *International Journal of Engineering Research and Technology (IJERT)*, vol. 9, pp. 497–500, 2020. [Online]. Available: <https://www.ijert.org/research/designing-a-smart-and-safe-drainage-system-using-artificial-intelligence-IJERTV9IS0.pdf>
- [6] K. Haripriya, S. S. Hammed, C. Preethi, S. Pavalarajan, R. Padmakumar, and M. M. Surya, "Automated drainage management system: Ai and iot-based blockage detection for smarter cities," in *2025 7th International Conference on Inventive Material Science and Applications (ICIMA)*. IEEE, 2025, pp. 998–1004. [Online]. Available: <https://doi.org/10.1109/ICIMA64861.2025.11073840>
- [7] S. H. Kwon and J. H. Kim, "Machine learning and urban drainage systems: State-of-the-art review," *Water*, vol. 13, no. 24, p. 3545, 2021. [Online]. Available: <https://doi.org/10.3390/w13243545>
- [8] P. S. Thanigaivelu, A. Sairam, P. Vijayan, A. S. Kumar, N. Mohankumar, and T. R. GaneshBabu, "Revolutionizing urban drainage: A smart iot approach to stormwater management using adaboosting algorithm," in *2024 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE)*. IEEE, 2024, pp. 1–6. [Online]. Available: <https://doi.org/10.1109/AMATHE61652.2024.10582217>
- [9] R. Sekita, T. Watanabe, T. Yoshizawa, and K. Nakanishi, "Reliability improvement of rainwater-drainage system using iot and ai," in *2024 Annual Reliability and Maintainability Symposium (RAMS)*. IEEE, 2024, pp. 1–6. [Online]. Available: <https://doi.org/10.1109/RAMS51492.2024.10457666>
- [10] S. Veerappan, "Edge-enabled smart storm water drainage systems: A real-time analytics framework for urban flood management," *Journal of Smart Infrastructure and Environmental Sustainability*, vol. 1, no. 1, pp. 52–59, 2024. [Online]. Available: <https://doi.org/10.17051/JSIES/01.01.06>

- [11] U. K. Sharma and A. Maurya, "Ai for predictive urban water drainage systems," *International Journal of Progressive Research in Engineering Management and Science*, vol. 4, no. 12, pp. 763–767, 2024. [Online]. Available: <https://www.doi.org/10.58257/IJPREMS37638>
- [12] P.-C. Huang and K. T. Lee, "An alternative for predicting real-time water levels of urban drainage systems," *Journal of Environmental Management*, vol. 347, p. 119099, 2023. [Online]. Available: <https://doi.org/10.1016/j.jenvman.2023.119099>
- [13] Y. Liu, Y. Liu, J. Zheng, F. Chai, and H. Ren, "Intelligent prediction method for waterlogging risk based on ai and numerical model," *Water*, vol. 14, no. 15, p. 2282, 2022. [Online]. Available: <https://doi.org/10.3390/w14152282>